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31 December 1975
SUPERSEDING
MIL-48441(MU)
31 January 1974

MILITARY SPECIFICATION

ELBOW ASSEMBLY, PASSIVE: 11733401

This specification is approved for use by all Department and Agencies of the Department of Defense*

1. SCOPE

1.1 This specification covers a modular unit of the Tank Commander's M36E1 Periscope or the Tank Gunner's M32E1 and M35E1 .Periscope, herein After referred to as Elbow Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel, General Specification Governing the Manufacture and Inspection.of |
| MIL-P-14232 | Parts, Equipment and Tools for Army Materiel, Packaging and Packing of |
| - MIL-I-45607 | Inspection Equipment, Acquisition, Maintenance and Disposition of |
| MIL-R-48577 | Regulator Assembly |
| MIL-I-49040 | image Intensifier Assembly, 25MM Micro-channel Inverter |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms aid Definitions |
| M1L-STD-81O | Environrnental Test Methods |

DRAWINGS

U.S. ArmY, Frankford Arsenal

| | |
|---|---|
| A7284901 | Dioptometer |
| F8565556 | Pressure Tester, Two-Station |
| F11729724 | Gage Interchangeability, Maximum |
| F11732100 | Gage Interchangeability, Minimum |



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| | |
|---|---|
| F11733401 | Elbow Assembly |
| D11738130 | Calibration Instructions For Elbow Assembly |
| F11746721 | Boot, Objective |
| F11746724 | Holder, .Eyepiece |
| C11746729 | Holder, Eyepiece 10″ Collimator |
| D11746730 | Eyepiece Assembly 10″ Collimator |
| F11746736 | Test Set-up for M35 Elbow Assembly |

PACKAGING DATA SHEET

| | |
|---|---|
| PS11733401 | Packaging of Elbow Assembly, Passive 11733401 |

(copies of specifications, standards, drawings and packaging data sheet required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Fabrication. - The elbow assembly shall be manufactured in accordance with Drawing F11733401 and drawings pertaining thereto,

3.2 Interchangeability. - The elbow assembly shall be capable of insertion into and removal from Periscope: M32E1, 35E1 and M36E1 (gages are specified in lieu of periscopes in 4.6.10).

3.3 General specifications. - The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926,

a. Order of precedence
b. Dimensions and tolerance
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Storage temperature. - The elbow assembly shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature (plus 60° to plus 90°F) after having been exposed and thermally stabilized at ambient temperatures of minus 80° and plus 160°F.

3.4.2 Operating temperatures. - The elbow assembly shall meet the applicable requirements of 3.5 while exposed and thermally stabilized at temperatures of minus 40°F and plus 150°F.

3.5 Operability.-



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3.5.1 Diopter movement.- The running torque required to rotate the diopter movement through the range specified in 3.6.4.7.1 shall be within the limits of 1.5 and 6.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 15 lb-ins.

3.5.2 Focusing mechanism.- The running torque required to rotate Ring 11733436 through the range specified in 3.6.9 shall be within the limits of 4.0 and 14.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 24.0 lb-ins.

3.5.3 Disengagement of boresight knob locks.- The direct force required to disengage each boresight knob from the lock position shall be within the limits of 2.0 to 5.0 pounds (lbs) when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the direct force shall not exceed 9.0 lbs.

3.5.4 Rotation of boresight knobs.- The running torque required to rotate each boresight knob through the range specified in 3.6.4.3 shall be within the limits of 0.5 and 5.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2 the running torque shall not exceed 8.0 lb-ins.

3.5.5 Shutter lever and on-off switch.- The torque required to rotate the shutter lever from the off to on position shall be within the limits of 4.5 to 6.5 lb-ins when checked at standard temperature. When checked at temperature conditions specified in 3.4.2, the torque shall not exceed 10 lb-ins.

3.5.6 Gain adjust and reticle brightness control.- This requirement shall be met at standard temperature and the temperature conditions specified in 3.4.2. Knob MS 91528-OE1B shall-be manually operable through the full limits of the rheostat.

3.6 Performance.

NOTE: The following references apply throughout this specification. Datum Plane “X” and Reference Line “Y” as shown on Sheet 1 of Drawing F11733401.

3.6.1 Optical preassembly characteristics.

3.6.1.1 With image tube removed.- Unless otherwise specified the requirements 3.6.1.2 to 3.6.1.4.2 inclusive shall be met with the image tube removed from the elbow assembly. “

3.6.1.2 Resolution - objective system.- The requirements of 3.6.1.2.1 and 3.6.1.2.2 shall be met with the specified line pairs per millimeter positioned at, and projected from the focal plane of the objective system.



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3.6.1.2.1 On axis.- The objective system shall resolve 10, 24, 32, 40, 48, 56, 64 and 72 line pairs per milliter on axis.

3.6.1.2.2 Off axis, 0.35 inch. - The objective system shall resolve 16, 24 32, 40, 48, 56, 64 and 72 line pairs per millimeter off axis 0.35 inch,

3.6.1.3 Resolution - eyepiece.- The requirements of 3.6.1.3.1 and 3.6.1.3.2 shall be met with the specified line pairs per millimeter positioned at and projected from the focal plane of the eyepiece.

3.6.1.3.1 On axis. - The eyepiece shall resolve 160 line pairs per millimeter on axis.

3.6.1.3.2 Off axis, 0.35 inch. - The eyepiece shall resolve 72 line pairs per millimeter Qff axis 0.35 inch. (Eyepiece may be refocused using the diopter adjustment ± .25 diopter).

3.6.1.4 Transmittance. - The requirements of 3.6.1.4.1 and 3.6,1.4.2 shall be met with a source of radiance at the focal plane of the system specified,

3.6.1.4.1 Objective system.

a. Luminous transmittance shall be not less than 72% when observed through any unobstructed area of the entrance pupil of the elbow assembly.

b. Transmittance at .85 micron shall be not less than 72% when observed as in 3.6.1.4.1a.

3.6.1.4.2 Eyepiece.- Luminous transmittance shall be not less than 82% when observed through the elbow assembly exit pupil.

3.6.2 Electrical pre-assembly characteristics.

3.6.2.1 Regulator. - The regulator shall have met the requirements of MIL-R-48577.

3.6.3 Elbow assembly-with Image tube installed.- Unless otherwise specified, the following tests shall be performed with the image tube energized. The voltage required to energize the image tube shall be supplied from a power source capable of supplying and maintaining the required voltage. The image tube shall be permitted a stabilization period of 15 seconds prior to conducting these tests.

3.6.3.1 Image intensifier tube. - The image intensifier tube shall be energized by the application of 24 VDC ± 6 VDC to Contact 11733497 or application of 3 VDC to Connector 11733493 (Ref, Dwg. F11733401).

3.6.3.2 Reticle projector lamp. - The reticle projector lamp shall be energized by the application of 24 VDC ± 6 VDC Contact 11733497 (Ref. Dwg. F11733401). The reticle brightness shall be adjustable from off



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to full bright by adjusting the reticle brightness control knob MS 91528-OE1B Clockwise. The reticle markings shall appear clearly defined while viewing through the eyepiece with the image tube energized.

3.6.3.3 Reticle brightness. - With a minimum of 18 and a maximum of 30 volts applied to the Contact 11733497 the reticle markings shall be visible when viewing through the eyepiece with the image intensifier tube energized (see 3.8.9.1). The reticle brightness shall be adjustable from off to full bright by adjusting the reticle brightness control Knob MS91528-OE1B clockwise.

3.6.3.4 Image steadiness.- There shall be no visible flicker, no movement of the image nor audible sparking when observed for 1 minute. This observation shall be made after the expiration of 15 seconds tube stabilization time.

3.6.3.5 On-Off switch.- The image intensifier tube shall shut down when 24 VDC is removed from Contact 11733497 with the on-off switch in the on position. The tuve shall not be energized by reapplication of the 24 VDC unless the on-off switch is recycled from off to on.

3.6.4 Collimation. - This requirement shall be met prior to meeting the requirements of 3.6.5.

3.6.4.1 Pre-Vibration/Pre-Shock setting.- An incident line of sight shall be provided which is perpendicular to Datum Plane “X”, within 0.3 mil. Coincidence of this line of sight with the elbow assembly reticle shall be established by rotation of the boresight knobs.

3.6.4.2 Reticle movement.- Reticle (boresight cross) movement shall be measured against an external target from the collimated setting of 3.6.4.1 to meet the requirements of 3.6.4.2.1 and 3.6.4.2.2.

3.6.4.2.1 Elevation. - Clockwise rotation of the elevation knob shall move the reticle, a minimum of 7.0 mils from the collimated setting. Counterclockwise rotation of the elevation knob shall move the reticle, a minimum of 7.0 mils from the collimated setting.

3.6.4.2.2 Deflection. - Clockwise rotation of the deflection knob shall move the reticle, a minimum of 7.0 mils from the collimated setting. Counterclockwise rotation of the deflection knob shall move the reticle, .a minimum of 7.0 mils from the collimated setting.

3.6.4.3 Knob rotation. - Continuous movement of the reticle shall be evident until limited by positive engagement of clockwise and counterclockwise stops in each boresight knob.

3.6.4.4 Line of sight vairiation.- Variation in the line of sight in elevation, as the deflection knob is rotated by at least 7.0 mils in each direction from the collimated setting of 3.6.4.1, shall not exceed 0.3 mil (total variation not to exceed 0.6 mil).

3.6.4.5 Reticle shift. - Disengagement or reversal of either knob shall not cause the line of sight to shift by more than 0.3 mil in a direction perpendicular to the intended motion.



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3.6.4.6 Boresight backlash.- Backlash in each boresight mechanism shall not exceed 0.3 mil.

3.6.4.7 Eyepiece focus. - The diopter scale index shall index zero within plus or minus 1/4 diopter when the screen is brought into best focus with the aid of a dioptometer.

3.6.4.7.1 Diopter movement. - The diopter movement shall be adjustable through a minimum range of -4.0 to +4.0 diopters from the zero diopter scale position of the eyepiece as read on a dioptometer when viewing the screen of the image tube,

3.6.4.8 Diopter backlash.- Backlash in the diopter movement shall not exceed 1/8 diopter.

3.6.4.9 Image tilt. - The image of a line originally parallel to Surface "X" and perpendicular to Reference Line "Y" shall emerge from the eyepiece parallel to the horizontal reticle line of an external telescope within 1.0 degree of arc.

3.6.4.10 Parallelism of reticle and image.- The image of a target line which is originally parallel to Surface "X" and perpendicular to Reference Line "Y" shall emerge from the eyepiece parallel to the horizontal reticle line within 1.0 degree of arc.

3.6.5 Vibration/Shock.

3.6.5.1 Vibration "A". - This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to the contact for the regulator and the reticle projector lamp (Reference Drawing F11733401). Vibration shall be applied along each of three mutually perpendicular axes of the elbow assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be .65 G (peak).

b. During a frequency sweep of 5 to 9 Hz, increase acceleration linearly from .65 to 1.5 G's (peak).

c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G's (peak).

d. During a frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5 G's (peak).

e. During a frequency sweep of 9 to 5 Hz. decrease acceleration linearly from 1.5 G's to .65 G (peak).

f. At 5 Hz, the acceleration shall be .65 G (peak).

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the elbow



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assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the elbow assembly shall meet the requirements of 3.2, 3.6 through 3.6.4.10 inclusive and 3.6.6 through 3.8 inclusive.

3.6.5.2 Vibration "B". - This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Reference Drawing F11733401) Vibration shall be applied along each of three mutually perpendiclar axes of the elbow assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be 1.0 G (peak).

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of 1.0 G (peak).

c. During a frequency sweep of 500 to 5 Hz, maintain constant acceleration of 1.0 G (peak).

d. At 5 Hz, the acceleration shall be 1.0 G (peak).

The frequency of applied vibration shall be sweept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minute ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes ± 3 minutes). There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the elbow assembly shall show no evidence of physical failure subsequent to vibration. Following the vibratio ntest, the elbow assembly shall meet the requirements of 3.2, 3.6 through 3.6.4.10 inclusive and 3.6.6 through 3.8 inclusive.

3.6.5.3 Shock. This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Reference Drawing F11733401). The elbow assembly shall be subjected to 24 cumulative -shocks consisting of three each of 100 G's, then 40 G's, applied in each direction along each of the two mutually perpendicular horizontal axes call four sides). The elbow assembly shall also be subjected to six cumulative shocks consisting of three each at 90 G's, then 35 G's applied in one direction along the vertical axis (normal operating position of elbow assembly only). The shock pulse shall have a half sine wave. For 100 G's ± 10% and 90 G's ± 10% the shock pulse shall have a time duration of 3 ± .2 mini-seconds; for 40 G's ± 10% and 35 G's ± 10% the shock pulse shall have a time duration of 18 ± 3 milliseconds. Subsequent to shock, there shall be no failure or flicker of light emission from the image tube nor the reticle projector lamp. The elbow assembly shall also show no evidence of physical failure. Following the shock test the elbow assembly shall meet the requirements of 3.2, 3.6 through 3.6.4.10 inclusive and 3.6.6 through 3.8 inclusive.

3.6.6 Post-Vibration/Post-Shock collimation.- The original collimation pre-vibration/pre-shock settings established in 3.6.4.1 shall not deviate more than 0.3 mil (including fixture error) due to vibration/shock, as observed through the eyepiece.



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3.6.7 Electron image shift.- Subsequent to vibration and shock, the elbow assembly shall be subjected to ten cycles of power on and power off at rated voltage; the voltage being applied for 15 seconds and removed for 10 seconds. The collimation of the elbow assembly at the expiration of each on period shall not deviate more than 0.3 mil.

3.6.8 Image centering.- An incident line of sight perpendicular to Datum Plane "X" (within 0.3 mil) shall appear centered in the eyepiece field of view within 2.5 degrees relative to an exit axis parallel to the datum plane specified above.

3.6.9 Objective focusing range.- Rotation of Ring 11733436 shall cause the objective to bring targets from 50 meters to 3000 meters into focus l

3.6.10 Resolution. - The elbow assembly shall be capable of resolving the following vaules (for the axis specified when observing a resolution target having the background luminance in foot lamberts (ft-1) and the contrast specified. Contrast (for the purpose of this requirement) is defined as the difference in density between the dark lines and the light background.

| RESOLUTION TARGET | | RESOLUTION |
|---|---|---|
| Background Luminance | Contrast (density difference between black and clear) | ON AXIS |
| 1 to 5 x 10-3 ft-1 | 2.0 or greater | 22 L-pr/mm |
| 6 x 10-5 ft-1 or less | 2.0 or greater | 4 L-pr/mm |

3.6.11 Luminance gain. - The luminance gain shall be no less than 625, when viewing an evenly illuminated field at 5 x $10^{-4}$ ft-lamberts with the gain control turned fully clockwise.

3.6.11.1 Gain control. - When viewing an evenly illuminated field at 5 x $10^{-4}$ ft-lamberts, the system gain shall vary as a function of the position of the gain control knob. Minimum gain shal be at the full counterclockwise position and increase steadily as the gain control varies to the full clockwise limit.

3.6.12 Field fo view. - The field of view shall be not less than 110 milliradians.

3.6.13 Projected reticle pattern.

3.6.13.1 Reticle image. - The image of the reticle pattern shall be well defined and clearly visible when viewed with the unaided eye under the following conditions:

a. The entrance window covered to prevent entrance of light,
b. The elbow assembly viewing an evenly illuminated field of not less than 1 to 5 x $10^{-3}$ foot lamberts luminance and the reticle voltage set at 18 VDC.



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3.6.13.2 Reticle accuracy.- The angular value between the boresight cross and the maximum range graduation of the projected reticle pattern shall be within 2.0 percent of the value specified on the reticle drawing.

3.7 Cleanliness.

a. Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures and adhesive separations on any glass component when viewing through either end.

b. Dirt. - When viewing through the eyepiece with the image intensifier tube energized, there shall be no more than fire particles visible on the screen surface. The size of any particle shall not exceed the -apparent width of the reticle line.

3.8 Sealing.- The elbow assebly shall shown no evidence of leakage in excess of 0.2 pound per square inch gage (psig) when subjected to an internal pressure of 5 ± 0.10 psig for a minimum of 2 hours. The interior atmosphere of the accepted elbow assembly shall contain nitrogen (having a dew point at least as low as minus 25°F at atmospheric pressure) between 0.25 and 0.50 Pound Per square inch gage (psig).

3.9 Reliability.- the elbow assembly shall have a mean time between failure (MTBP) of 2000 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his won facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspectio terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approvel.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sanple for first article approval test shall consist of three elbow assemblies plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing shall be obtained from the same source 0f supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it



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conforms to all the requirements of the contract and submit a record of the inspection With the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates to conformance for materials, shall be submitted to the Government for approval. The. Government reserVes the right to witness the contractor's inspection.

4.3 Inspection provisions;

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot SiZe, lot formation and presentation of lots shall be in accordance with "submission of Product" previsions of MIL-S-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and sub-assemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-P-13926 shall apply.

4.3.2.1.1 Certification. - Prior to assembly as a complete elbow assembly, the objective assembly, eyepiece assembly, image intensifier tube and regulator shall be certified to the requirements and tests in Table I. Certification shall include test data and results for the characteristics specified in Table I. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

TABLE I

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 301. | Resolution, objective system, on axis | 3.6.1.2.1 | 4.6.2.1 |
| 302. | Resolution, objective system, off axis, 0.35 inch | 3.6.1.2.2 | 4.6.2.2 |
| 303. | Resolution, eyepiece, on axis | 3.6.1.3.1 | 4.6.2.4 |
| 304. | Resolution, eyepiece, off axis, 0.35 inch | 3.6.1.3.2 | 4.6.2.5 |
| 305. | Transmittance, objective system | 3.6.1.4.1 | 4.6.2.3 |
| 306. | Transmittance, eyepiece | 3.6.1.4.2 | 4.6.2.6 |
| 307. | Regulator tests | 3.6.2.1 | 4.6.1.1 |



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4.3.2.2 Elbow Assembly*

4.3.2.2.1 First article. - First article testing shall be conducted in accordance with 4.3.2.1 and Tables I through VI and shall also meet the MTBF requirement and test in 3.9 and 4.6.9 respectively.

4.3.2.2.2 100% Inspection. - The requirements and related tests in Table II shall be subjected to 100% inspection.

TABLE II. 100% INSPECTION

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 1. | Interchangeability | 3.2 | 4.6.10 |
| 2. | Image intensifier tube | 3.6.3.1 | 4.6.6.1 |
| 3. | Reticle projector lamp | 3.6.3.2 | 4.6.6.2 |
| 4. | Reticle brightness | 3.6.3.3 | 4.6.6.3 |
| 5. | On-Off switch | 3.6.3.5 | 4.6.6.4 |
| 6. | Line of sight variation | 3.6.4.4 | 4.6.6.12 |
| 7. | Reticle shift | 3.6.4.5 | 4.6.6.13 |
| 8. | Eyepiece focus | 3.6.4.7 | 4.6.6.15 |
| 9. | Image centering | 3.6.8 | 4.6.6.7 |
| 10. | Resolution | 3.6.10 | 4.6.6.20 |
| 11. | Luminance gain | 3.6.11 | 4.6.6.21 |
| 12. | Gain control | 3.6.11.1 | 4.6.6.22 |
| 13. | Sealing | 3.8 | 4.6.8 |

NOTE: These tests shall be conducted at a temperature between +60° and +90F.

4.3.2.2.3 Sampling. - Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and Table III below. Examinations and tests for packaging, packing and working shall be in accordance with MIL-P-14232 and Section 5 herein. In addition to 100% inspection as specified in Table II, the tabulated characteristics i Tables I and III through VI shall constitute the minimum inspection to be perfomed by the supplier after First Article approval and prior to Government acceptance or rejection by item or lot.

TABLE III. - CLASSIFICATION OF DEFECTS

CLASS
CRITICAL: NONE
MAJOR: AQL 0.65% DEFECTIVE

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 101. | Operability, diopter movement | 3.5.1 | 4.6.4.1 |
| 102. | Operability, focusing mechanism | 3.5.2 | 4.6.4.2 |
| 103. | Operability, disengagement of boresight knob locks | 3.5.3 | 4.6.4.3 |
| 104. | Operability, rotation of boresight knobs | 3.5.4 | 4.6.4.4 |
| 105. | Operability, shutter lever and On-Off switch | 3.5.5 | 4.6.4.5 |



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| 106. | Operability, gain adjust and reticle brightness control | 3.5.6 | 4.6.4.6 |
| 107. | Image steadiness | 3.6.3.4 | 4.6.6.5 |
| 108. | Collimation, Pre-Vibration setting | 3.6.4.1 | 4.6.5.1, 4.6.6.9 |
| 109. | Reticle movement, elevation | 3.6.4.2.1 | 4.6.6.10.1 |
| 110. | Reticle movement, deflection | 3.6.4.2.2 | 4.6.6.10.2 |
| 111. | Knob rotation | 3.6.4.3 | 4.6.6.11 |
| 112. | Boresight backlash | 3.6.4.6 | 4.6.6.14 |
| 113. | Diopter movement | 3.6.4.7.1 | 4.6.6.16 |
| 114. | Diopter backlash | 3.6.4.8 | 4.6.6.17 |
| 115. | Image tilt | 3.6.4.9 | 4.6.6.6 |
| 116. | Parallelism of reticle and image | 3.6.4.10 | 4.6.6.18 |
| 117. | Vibration "B" | 3.6.5.2 | 4.6.5.2 |
| 118. | Post-Vibration collimation | 3.6.6 | 4.6.5.3 |
| 119. | Post-Vibration examination | 3.6.5.2 | 4.6.5.4 |
| 120. | Electron image shift | 3.6.7 | 4.6.6.19 |
| 121. | Objective focusing range | 3.6.9 | 4.6.6.8 |
| 122. | Field of view | 3.6.12 | 4.6.6.23 |
| 123. | Projected reticle pattern, reticle image | 3.6.13.1 | 4.6.6.24.1 |
| 124. | Projected reticle pattern, reticle accuracy | 3.6.13.2 | 4.6.6.24.2 |
| 125. | Cleanliness, dirt | 3.7b | 4.6.6.25 |
| 126. | Cleanliness, defects | 3.7a | 4.6.7 |

MINOR: NONE

NOTE: The tests in Table III shall be conducted at temperature between +60° and +90°F.

4.3.2.2.4 Environmental.- Three elbow assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III.

TABLE IV

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 308. | Storage Temperature (-80° to +160°F) | 3.4.1 | 4.6.3.1 |
| 309. | Operating Temperature (-40° to +150°F) | 3.4.2 | 4.6.3.2 |
| 310. | Operability, diopter movement | 3.4.2, 3.5.1 | 4.6.3.2, 4.6.4.1 |
| 311. | Operability, focusing mechanism | 3.4.2, 3.5.2 | 4.6.3.2, 4.6.4.2 |
| 312. | Operability, Disengagement of boresight knob locks | 3.4.2, 3.5.3 | 4.6.3.2, 4.6.4.3 |
| 313. | Operability, rotation of boresight knob | 3.4.2, 3.5.4 | 4.6.3.2, 4.6.4.4 |
| 314. | Operability, shutter lever and on-off switch | 3.4.2, 3.5.5 | 4.6.3.2, 4.6.4.5 |
| 315. | Operability, gain adjust and reticle brightness control | 3.4.2, 3.5.6 | 4.6.3.2, 4.6.4.6 |


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4.3.3 Control tests. - Three elbow assemblies shal be selected and meet the requirements and tests of Table V. These tests shall be conducted as follow-up tests to the first article samples at 25 percent, 50 percent and 75 percent completion of the production contractc

TABLE V

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 316. | Collimation, Pre-Vibration setting | 3.6.4.1 | 4.6.5.1, 4.6.6.9 |
| 317. | Vibration "A" | 3.6.5.1 | 4.6.5.5.1 |
| 318. | Post-Vibration collimation | 3.6.6 | 4.6.5.3 |
| 319. | Post-Vibration examination | 3.6.5.1 | 4.6.5.4 |
| 320. | Collimation Pre-Shock setting | 3.6.4.1 | 4.6.5.1, 4.6.6.9 |
| 321. | Shock | 3.6.5.3 | 4.6.5.5.2 |
| 322. | Post-Shock collimation | 3.6.6 | 4.6.5.3 |
| 323. | Post-Shock examination | 3.6.5.3 | 4.6.5.4 |

4.3.4 General. - One.elbow assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table VI.

TABLE VI

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 324. | Fabrication | 3.1 | Applicable Drawing - Visual |
| 325. | General specifications | 3.3 | MIL-F-13926 - Visual |

NOTE: These tests shall be conducted at a temperature between +60° and +90°F.

4.4 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Inspection equipment.- Except as otherwise provided for by hte contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Governmant design. - All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

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4.5.2.2 Contractor design- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Prodedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is entended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Special inspection equipment.

4.5.2.3.1 Sealing. - Special testing equipment to test sealing of the elbow assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station,

4.5.2.3.2 Performance. - Special test equipment to determine performance of the elbow assembly shall conform to the requirements of Drawing F11746736 (Test Set-Up for M35 Elbow), Drawing F11746724 (Holder, eyepiece), Drawing F11746721 (Boot, Objective), Drawing D11746730 (Eyepiece) Assembly, 10" Collimator), Drawing c11746729 (Holder, Eyepiece, 10" Collimator) and drawings pertaining thereto.

4.5.2.3.2.1 Interchangeability. - Maximum and Minimum interchangeability gages shall conform to Drawings F11729724 and F11732100 respectively,

4.6 Test methods and procedures,

4.6.1 Electrical pre-assembly characteristics.

4.6.1.1 Regulator Assembly, MIL-R-48577, pre-installation.- Each regulator assembly shall be tested to the requirements of MIL-R-48577. prior to insertion into the elbow assembly to determine its proper functioning prior to energizing the image intensifier tube as specified in 3.6.2.1,

4.6.2 Optical pre-assembly characteristics, with image tube removed.- The following tests of 4.6.2.1 through 4.6.2.6 inclusive, shall be performed with Eyepiece Assembly 11733402 and image intensifier tube removed from the elbow assembly. The test results for all characteristics shall be recorded and be available to the Government.

4.6.2.1 Resolution - objective system - on axis.- This test shall be performed utilizing the test equipment conforming to Drawing F11746736 (Test Set-Up forM35 Elbow). With Adapter Prism "M" removed, the elbow objective assembly (with Boot Objective 11746721 installed) shall be placed o Adapter "H" and energized by power pack "F", The Holder, Eyepiece 10" Collimator 11746729 shall be positioned on Collimator "G"



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after lamp "X" has been removed. The center pattern of the target shall be resolved when viewing through Eyepiece Assembly 11746730 positioned in the center slot of the eyepiece holder to determine compliance with 3.6.1.2.1.

4.6.2.2 Resolution - Objective system - off-axis, 0.35 inch.- This test shall. be Performed utilizing the test equiPment sPecified in 4.6.2.1, except that the Eyepiece Assembly 11746730 shall be positioned in the off-center slot of Holder, Eyepiece 11746729. The pattern in each of the four meridians shall be resolved when viewing through the eyepiece assembly to determine compliance with 3.6.1.2.2.

4.6.2.3 Transmittance - objective system.- This test shall be performed utilizing the test equipment specified in 4.6.2.1. With Collar, "U" slid out of the way, position Penta Prism "S" on a tool-maker's knee between the objective lens and Collimator "G" so that the line of sight is projected 90°. A photometer having a 7 millimeter aperture and appropriate filters (eyecorrection and 0.85 micron), shall be positioned such that it intercepts the line of sight from the penta prism. The light source shall present a uniform target luminance greater than 100 foot lamberts by means of a tungsten lamp held at a constant temperature of 2870° ± 100°K. While scanning the clear portion of the objective aperture, luminance transmittance shall be not less than the value specified in 3.6.1.4.1a when measured with the photometer, eye-corrected; transmittance at 0.85 micron shall be not less than the value specified in 3.6.l.4.lb when measured with the photometer filtered for 0.85 micron. The percentage of transmittance shall be computed uning the following formula:

$$\% \text{ Transmittance} = \frac{\text{Radiation of source as seen through Objective Assembly and Penta Prism}}{\text{Radiation of Source (measured through Penta Prism)}} \times 100$$

4.6.2.4 Resolution - eyepiece - on axis.- This test shall be performed utilizing test equipment conforming to Drawing F11746724 (Holder, Eyepiece). After positioning the eyepiece assembly in the eyepiece holder view through the eyepiece assembly and resolve the center pattern of the target to determine compliance with 3.6.1.3.1.

NOTE: An observation telescope of at least three power may be used as an aid to conduct this test and the test of 4.6.2.5.

4.6.2.5 Resolution - eyepiece - off axis, 0.35 inch. - This test shall be performed utilizing the test equipment specified in 4.6.2.4. While viewing through the eyepiece assembly the pattern in each of the four meridians shall be resolved to determine compliance with 3.6.1.3.2. Any adjustment of the eyepiece diopter adjustment ring to obtain best focus of the resolution target shall not exceed ± .25 diopters to determine overall compliance with 3.6.1.3.2.

4.6.2.6 Transmittance - eyepiece.- This test shall be performed utilizing the test equipment specified in 4.6.2,4. An eye-corrected



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photometer having a 7 millimeter aperture and appropriate filters, shall be positioned such that it intercepts the line of sight from the exit lens of the eyepiece. The light source shall present a uniform target luminance greater than 100 foot lamberts, by means of a tungsten lamp held at a constant temperature 2870° ± 100°K. The transmittance measurements shall be made with the eyecorrected photometer, and shall not be less than the value specified in 3.6.1.4.2. The percentage of transmittance shall be computed using the following formula:

$$\% \text{ Transmittance} = \frac{\text{Luminance of source as seen through Eyepiece Assembly} \times 100}{\text{Luminance of source}}$$

4.6.3 Environmental.

4.6.3.1 Storage temperatures.- This test shall be performed on elbow assemblies procured as a repair part item. When the elbow assembly is provided with a complete periscope this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The elbow assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE below) to -80°F. The elbow assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see NOTE below) to +160°F. The elbow assembly shall remain at this temperature for a "minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the elbow assembly removed from the. test chamber. The elbow assembly shall then be subjected to the visual, tactile, and audible examination of 4.6.5.4. The elbow assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.3.1 and 4.6.3.2.

4.6.3.2 Operating temperatures.- This test shall be performed on elbow assemblies procured as a repair part item. When the elbow assembly is provided with a complete periscope this temperature test will not be required when the complete periscope is subjected to temperature cycling. The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The elbow assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE in 4.6.3.1) to -40°F. The elbow assembly shall remain at this temperature for a minimum of 5 hours. t the completion of this 5 hour period and while at -40°F, the elbow assembly shall. be subjected to the tests in 4.6.4 to determine compliance with the applicable requirements of 3.5, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (See NOTE in 4.6.3.1) to +150°F. The elbow assembly shall remain at this temperature for a period of 5 hours.



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At the completion of the 5 hour period and while at +150°F, the elbow assembly shall again be subjected to the tests in 4.6.4 to determine compliance with the applicable requirements of 3.5, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see NOTE in 4.6.3.1) to room ambient temperature (+60° to +90°F) and the elbow assembly removed from the test chamber~

4.6.4 Operability.

4.6.4.1 Diopter movement.- Standard torque measuring equipment capable of measuring the torque values specified in 3.5.1 shall be utilized for this test. An appropriate adapter, to accomodate the diopter knob shall be used with the torque measuring equipment. Torque shall be meassured while rotating the diopter movement through the range specified in 3.6.4.7.1. The running torque required to rotate the diopter knob shall conform to the applicable requirements of 3.5.1 for the temperatures indicated.

4.6.4.2 Focusing mechanism.- Standard torque measuring equipment capable of measuring the torque values specified in 3.5.2 shall be utilized for this test. An appropriate adapter, to accommodate Ring " 11733436 shall be used with the torque measuring equipment. Torque shall be measured while rotating the ring through the range specified in 3.6.9. The running torque required to rotate Ring 11733436 shall conform to the applicable requirements of 3.5.2 for the temperatures indicated.

4.6.4.3 Disengagement of boresight knob locks.- Standard measuring equipment capable of measuring the force values specified in 3.5.3 shall be utilized for this test. The force required to disengage each boresight knob from the lock position shall conform to the applicable requirements of 3.5.3 for the temperatures indicated.

4.6.4.4 Rotation of boresight knobs.- Standard torque measuring equipment capable of measuring the torque values specified in 3.5.4 shall be utilized for this test. An appropriate adapter, to accommodate the boresight knobs shall be used with the torque measuring equipment. Torque shall be measured while rotating each boresight knob through the range specified in 3.6.4.3, The running torque required to rotate each boresight knob shall. conform to the applicable requirements of 3.5.4 for the temperatures indicated.

4.6.4.5 Shutter lever and on-off switch.- Standard torque measuring equipment capable of measuring the torque values specified in 3.5.5 shall be utilized for this test. The torque required to rotate the shutter lever from the off to on position and then back to the off position shall conform to the applicable requirements of 3.5;5 for the temperatures indicated.

4.6.4.6 Gain adjust and reticle brightness control.- Manually operate the Knob MS 91528-0E1B. The knob shall meet the requirements specified in 3.5.6, at the temperatures indicated in 3.4.2 in addition to the standard ambient temperature (+60° to +90°F). Elbow assemblies which have passed the above tests of 4.6.4.1 through 4.6.4.6 inclusive shall be subjected to the tests in 4.6.5.



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4.6.5 Vibration/Shock.

4.6.5.1 Collimation, Pre-Vibration/Pre-Shock setting.- Prior to the Vibration/Shock tests, the elbow assembly shall be positioned and oriented as in 4.6.6 to insure conformance to 3.6.4. While viewing through the elbow assembly eyepiece, obtain coincidence of the elbow assembly reticle boresight cross with the projected target image cross-lines, by means of the boresight knobs. When coincidence has been obtained, slip the scales of the boresight knobs to a setting of (4) and recheck to insure coincidence is not disturbed.

NOTE: Insure that knob scales are not disturbed (re-slipped) from their original setting of (4) throughout remainder of tests~

4.6.5.2 Vibration "B". This test shall be performed on elbow assemblies procured as a repair part item. When the elbow assembly is provided with a complete periscope this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the elbow assembly shall be subjected to the test procedure specified in 4.6.5.1 to determine compliance with 3.6.4. Subsequent to the test of 4.6.5.1, the elbow assembly shall be vibrated in accordance with Test Method 514.2, procedure VIII (Tracked Vehicles) of MIL-STD-81O, except the test level and frequency range shall be as specified in 3.6.5.2. Vibration shall be applied along each of three mutually perpendicular axes of the elbow assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-81O. The vibrating equipment shall also be capable of vibrating the elbow assembly along three mutually perpendicular axes while the elbow assembly is maintained in its normal mounting or operating position (as depicted in the-front view on Sheet 1 of Drawing F11733401). The elbow assembly shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as Periscope: M32E1, M35E1 and M36E1. Prior to vibration, the image intensifier tube and the reticle projector lamp shall be energized in accordance with the 3.6.5.2. After allowing a 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the elbow assembly eyepiece. The elbow assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.6.5.2. During each vibration sweep cycle, the screen surface of the tube and the reticle pattern shall be monitored to assure that there is no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp. Immediately following the vibration cycle, and prior to the examination for-physical defects resulting from poor workmanship and materials, the elbow assembly shall be subjected to the test in 4.6.5.3 and then to the test in 4.6.5.4 Elbow assemblies that have passed the above tests shall then be subjected to the tests specified in 4.6.6, 4.6.7 and 4.6.8.

4.6.5.3 Post-Vibration/Post-Shock collimation.- Subsequent to Vibration/Shock test, the elbow assembly shall be repositioned as in



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4.6.5.1. Do not disturb the boresight knob scale settings of (4) during transfer of the elbow assembly. Observe through the elbow assembly eyepiece and adjust by means of the boresight knobs to obtain coincidence of the elbow reticle boresight cross with the target image. Boresight knob adjustments shall not exceed the tolerance specified in 3.6.6, as measured on the boresight knob scales.

4.6.5.4 Post-Vibration/Post-Shock examination.- Subsequent to the test in 4.6.5.3, the elbow assembly shall be thoroughly examined for physical failure resulting from poor workmanship and materials. Prior to being subjected to the remaining tests specified herein, a visual examination shall be made for missing or damaged parts. With the exception of the slip scales, a tactile examination shall be made for functioning of moving parts (boresight knobs, boresight knob scales, etc.) for undue irregularities of movement, friction or looseness. An audible examination shall be made of the elbow assembly for loose internal optical and mechanical components, any one of the possible defects defined above shall be cause to consider an elbow assembly defective,

4.6.5.5 Vibration "A''/Shock.- In addition to first article sample, the tests in 4.6.5.5.1 (Vibration "A") and 4.6.5.5.2 (Shock) shall be performed on three elbow assemblies randomly selected at 25%, 50% and 75% completion of the production contract.

4.6.5.5.1 Vibration "A". - This test shall be performed on elbow assemblies procured as a repair part item. When the elbow assembly is provided with a complete periscope this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the elbow assembly shall be subjected to the test procedure specified in 4.6.5.l to determine compliance with 3.6.4. Subsequent to the test of 4.6.5.1, the elbow assembly shall be vibrated in accordance with Test Method 514.2, Procedure VIII (Tracked Vehicles) of MIL-STD-810, except the test level and frequency range shall be as specified in 3.6.5.1. Vibration shall be applied along each of three mutually perpendicular axes of the elbow assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above-shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the elbow assembly along three mutually perpendicular axes while the elbow assembly is maintained in its normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F11733401). The elbow assembly shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as periscope: M32E1, M35E1 and M36E1. Prior to vibration, the image intensifier tube and the reticle projector lamp shall be energized in accordance with 3.6.5.1. After allowing a 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the elbow assembly eyepiece. The elbow assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.6.5.1. During each vibration sweep



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cycle, the screen surface of the tube and the reticle pattern shall be monitored to assure that there is no failure or flicker of light emission . from the image intensifier tube nor the reticle projector lamp. Immediately following the vibration cycle, and prior to the examination for physical. defects resulting from poor workmanship and materials, the elbow assembly shall be subjected to the test in 4.6.5.3 and then to the test. in 4.6.5.4. Elbow assemblies that have passed the above tests shall then be subjected to the tests specified in 4.6.6, 4.6.7 and 4.6.8.

4.6.5.5.2 Shock.- This test shall be performed on elbow assemblies procured as a repair part item. When the elbow assembly is provided with a complete. periscope this shock test will not be required when the complete periscope-is subjected to shock. Prior to performing this test, the elbow assembly. shall be subjected to the test procedure specified in 4.6.5.1 to determine compliance with 3.6.4. The shock equipment and . test procedure shall be in accordance with Procedure I, of Method 516.2 specified in MIL-STD-81O, except conditions such as "G" values, time duration and wave form shock pulse shall be as specified in 3.6.5.3. The elbow assembly shall be clamped on the shock equipment by means of an appropriate adapter having the same interface features as Periscope: M32E1, M35E1 and M36E1. Prior to shock, the image intensifier tube and reticle projector lamp shall be energized in accordance with 3.6.5.3. After allowing a 15-second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the elbow assembly eyepiece. The elbow assembly shall then be shocked in accordance with the test procedure specified above, except the application of shock along the vertical axis shall be in one direction only (normal operating position . of the elbow assembly). Subsequent to each shock, the screen surface of the tube and the reticle pattern shall be observed. During these observations> there shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp. Immediately following the shock test, and prior to the examination for physical defects resulting from poor workmanship and materials the elbow assembly shall be subjected to the test in 4.6.5.3 and then to the test in 4.6.5.4. Elbow assemblies that have passed the above test shall then be subjected to the tests specified in 4.6.6, 4.6.7 and 4.6.8.

4.6.6 Performance, elbow assembly with image tube installed. - The following tests shall be performed for compliance with 3.6 and 3.6.3. . The elbow assembly shall be mounted on special testing equipment conforming to the requirements of 4.5.2.3.2. Set-up of the test equipment, together with the mounting and securing of the elbow assembly on the test equipments shall be performed in accordance with the instructions contained on the test equipment drawing. Unless otherwise specified during these tests, Collimator "G" shall be adjusted for infinity; the image intensifier tube and reticle projector lamp shall be energized in accordance with 3.6.3.1 and 3.6.3.2; the Focusing Ring 11733436 shall be adjusted for best focus of the target image after the diopter adjustment ring of the eyepiece is set for best focus of the phospher grain on the image intensifier tube; and Holder "T" containing Telescope "J" with Special Reticle "L" shall be removed from the optical line of sight of the test equipment. The image . intensifier tube shall be permitted a stabilization period of 15 seconds prig to conducting the optical tests.



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4,6.6.1 Image intensifier tube.- Apply 24 Vdc ± 6 Vdc to contact 11733497 (reference Drawing F11733401) and cycle the on-off switch from off to on by means of the shutter lever. After allowing a 15 second tube stabilization period, view through the elbow assembly eyepiece and observe the image tube screen, The screen surface of the tube shall be illuminated. Upon satisfactory completion of this portion of the test, remove the 24 VDC ± 6 VDC from connector 11733497 and apply 3 VDC to connector 11733493 (reference Drawing 11733401.). Cycle the on-off switch from off to on. After allowing a 15 second tube stabilization period, view through the elbow assembly eyepiece. The screen surface of the tube shall be illuminated to determine overall compliance with 3.6.3.1.

4.6.6.2 Reticle projector lamp.- Perform this test concurrent with the test in 4.6.6.1. With the voltage specified in 3.6.3.2 applied to contact 11733497 (reference Drawing F11733401) and the image intensifier tube energized, rotate the reticle brightness control Knob MS 91528-0E113 to the off position. Then while viewing through the elbow assembly eyepiece, rotate the control knob clockwise throughout its range of movement. During this clockwise rotation of the control knob, the reticle brightness shall be adjustable from off to full bright and the reticle markings shall appear clearly defined when observed through the elbow assembly eyepiece to determine compliance with 3.6.3.2.

4.6.6.3 Reticle brightness.- Perform this test concurrent with the test in 4.6.6.1. With each voltage specified in 3.6.3.3 applied to Contact 11733497 (Reference Drawing F11733401) and the image intensifier tube energized, rotate the reticle brightness control knob MS 91528-OE1B to be off position. Then while viewing through the elbow assembly eyepiece, rotate the control knob clockwise throughout its range of movement. During this clockwise rotation of the control knob, the reticle brightness shall be adjustable from off to full bright and the reticle markings shall appear clearly defined at each voltage setting when observed through the elbow assembly eyepiece to determine compliance with 3.6.3.3.

4.6.6.4 On-off switch. - This test may be performed at the same time the test of 4.6.6.1 is performed. With the voltage specified in 4.6.6.1 applied to Contact 11733497. view through the elbow assembly eyepiece and observe the illuminated screen surface of the tube. Then while the on-off switch is in the on position, remove the voltage from the contact and reapply. During the above observation, the screen surface of the tube shall shut off when voltage is removed and shall not illuminate by reapplication of the voltage unless the on-off switch is recycled from off to on to be in conformance with 3.6.3.5.

4.6.6.5 Image steadiness. - This test and the following tests of 4.6,6.6 and 4.6.6.7 shall be performed with the reticle projector lamp off. Holder “T” containing Telescope “J” with Special Reticle “L” shall be inserted in the slot against the pins provided in the main plate at the eyepiece area of the elbow assembly. With the aid of the telescope,



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view through the eyepiece of the elbow assembly and observe the telescopes reticle pattern and the target image for a time period of not less than one minute. During this one minute time period there shall be no visible flicker, no movement-of the image nor audible sparking as specified in 3.6.3.4. Image movement shall be determined bY means of the target. image with respect to the telescope reticle pattern. Sparking shall be determined by means of an audio examination.

4.6.6.6 Image tilt. - Perform this test concurrent with the test. in 4.6.6.5. With aid of the telescope, view through the eyepiece of the elbow assembly and establish one extremity of the telescope vertical reticle line coincident, to the vertical target image line. The opposite extremity of the targets vertical line shall be within the gate provided on the telescope reticle to be in compliance with 3.6.4.9.

4.6.6.7 Image centering. - Perform this test concurrent with the test in 4.6.6.6. With the aid of the telescope, view through the eye-piece of the elbow assembly and observe the projected target image. The . line of sight through the intersecting point of the vertical and horizon-tal target line image must be within, or in coincidence with the circle on the telescope reticle to appear centered in the eyepiece field of view within the value specified in 3.6.8. Upon completion of this test energize the reticle projector lamp.

4.6.6.8 Objective focusing range.- This portion of the test shall be performed with the 10″ collimator adjusted for 50 meters. With the diopter ring adjusted for best focus of the phosphor grain on the screen of the image tube, rotate focusing ring of the elbow assembly for best focus of target image to determine cornpli.ante with 3.6.9. Upon satis-factory completion of this portion of the test, the 10″ collimator shall be readjusted for infinity to determine compliance-with the 3000 meters specified in 3.6.9. With the diopter ring adjusted for best focus of the phosphor grain on the screen of the image tube, again rotate the focusing ring of the elbow assembly for best focus of the target image to determine overall compliance with 3.6.9.

4.6.6.9 Collimation, pre-vibration/Pre-Shock setting. - This test shall be conducted in accordance with 4.6.5.1 to determine compliance with 3.6.4.1.

4.6.6.10 Reticle movement.

4.6,6.10.1 Reticle movement, elevation.- Establish coincidence of the elbow assembly reticle boresight cross and the projected target image crosslines by means of the boresight knobs (do not slip knob scales). While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall move as specified in 3.6.4.2.1 as observed and measured on the target. Reestablish coincidence of the reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall move as specified in 3.6.4.2.1 as observed and measured on the target.



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4.6.6.10,2 Reticle movement, deflection.- Perform this test utilizing the procedure in 4.6.6.10.l except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) movements shall meet the requirements specified in 3.6.4.2.2 as observed and measured on the targets

4.6.6.11 Knob rotation. - Perform this test in conjunction with the tests in 4,6.6.10.1 and 4.6.6.1O.2. During rotation of each boresight knob (elevation and deflection), the boresight cross (intersecting point) shall be observed and show no evidence of irregularities (jump; reeep or hangup) in direction of travel until limited by engagement of each knob stop (clockwise and counterclockwise to be in compliance with 3.6.4.3. While the boresight knobs are at each stop position, back off the knobs for at least l/4-rotation and then again return the knobs to the stop. Perform this a minimum of 3 times. During these rotations, the knobs shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid, such that a distinct metallic click can be heard to verify that reticle travel is being limited only by means of the knob stops and not by means of contact with adjacent components to determine overall compliance with 3.6.4.3.

4.6.6.12 Line of sight variation.- Perform this test in a manner similar to the test in 4.6.6.10.2. During rotation of the deflection knob, any variation in elevation in the line of sight, through the bore-sight cross (intersecting point), shall not exceed the values specified in 3.6.4.4.

4.6.6.13 Reticle shift. - With backlash removed for direction of travel, establish coincidence of the elbow assembly reticle boresight cross as in 4.6.6.10.1. While observing the intersecting point of the elbow assembly reticle boresight mark, disengage the elevation boresight knob from the lock position and proceed to rotate the knob (insure back-lash is not introduced) for at least 1/4 turn, then reverse rotation of the knob (backlash included) for at least 1/4 turn. During the disengage-ment or reversal of the boresight knob, the line of sight through the elbow assembly reticle boresight mark shall not shift (jump) in a direc-tion perpendicular to the intended motion more than the tolerance specified in 3.6.4.5. In addition to checking shift at this setting, repeat the above at the midpoint and maximum point of travel in each direction for a total 5 checks overall. These 5 checks shall also be conducted with the deflection knob in compliance with 3.6.4.5.

4.6.6.14 Boresight backlash. - While viewing through the elbow assembly eyepiece, disengage and rotate the boresight knobs (insure backlash is removed for direction of travel) until a line of sight, through the intersection of the elbow assembly reticle boresight mark is positioned coincident with the intersecting point of the target image crosslines. With the boresight knobs in the lock position, record the boresight knob scale readings and the position of the line of sight. Disengage the boresight knobs and continue rotation approximately 1/2 of a revolution then return and engage the knobs to their original lock position as recored above. View through the eyepiece and record the



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position of the line of sight through the reticle foresight mark. Any difference between the recorded positions of the line of sight for each boresight knob is the amount of backlash. Backlash in each boresight mechanism shall not exceed the tolerance specified in 3.6.4.6, as measured on the target.

NOTE: Should overtravel occur during rotation of the boresight knobs, the test in nullified and shall be repeated.

4.6.-6.15 Eyepiece focus.- A calibrated diopometer conforming to Drawing A7284901 shall be utilized for measuring the accuracy of the zero diopter setting of the diopter scale. Set the diopometer objective scale to zero and adjust the dioptometer “eyepiece for best focus of the dioptometer reticle. Set the diopter knob of th eelbow assembly eyepiece to indicate zero diopter. Position the dioptometer at the eyepiece of the elbow assembly. While viewing through the dioptometer, obtain best focus of the phosphor grain on the image tube by adjustment of the diop-tometer objective scale. The phosphor grain of the image tube shall be in best focus within the tolerance specified in 3.6.4.7 as measured on the dioptometer objective scale.

4.6.6.16 Diopter movement.- This test shall be performed in con-junction with 4.6.6.15 to determine compliance with 3.6.4.7.1. Rotate the diopter adjustment ring from the established best focus of the target image to its clockwise stop and note the diopter setting on the scale; then rotate the diopter ring to its counterclockwise stop, note diopter setting on the scale. The diopter movement shall meet the requirements specified in 3.6.4.7.1 as measured o the diopter scale of the elbow assembly.

4.6.6.17 Diopter backlash.- With backlash excluded for direction of travel, rotate the diopter ring clockwise and obtain best focus of the phosphor grain on the screen of the image tube. Record the diopter scale reading. Continue rotation of diopter ring to its stop, then with backlash included, rotate the knob counterclockwise and obtain best focus of the phosphor grain on the screen of the image tube without overtravel. Record the diopter scale reading. The difference between the first and second recorded scale reading is the amount of backlash in the diopter movement. Backlash in the diopter movement shall not exceed the tolerance specified in 3.6.4.8.

NOTE: Should overtravel occur during rotation of the diopter ring the test is nullified and shall be repeated.

4.6.6.18 Parallelism if image and reticle.- Perform this test while observing through the elbow assembly eyepiece with the unaided eye. Position one extremity of the elbow assembly vertical reticle line coincident to the vertical target image line. The opposite extremity of the elbow assembly vertical reticle line shall be within the gate provided on the target image to be in compliance with 3.6.4.10.

4.6.6.19 Electron image shift.- Observe through the elbow assembly eyepiece and establish coincidence of the elbow assembly reticle bores cress and the projected target image crosslines by means of the boresight



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knobs. When coincidence has been established, proceed with the cycling as described in 3.6.7. During each on cycle, the projected target image with respect to the elbow assembly reticle shall be observed for repeatability of the coincident setting established above. At the expiration of each on cycle, any adjustment of the boresight knobs to reestablish coincidence of the reticle boresight cross and the projected image crosslines shall not exceed the value specified in 3.6.7, as measured on the boresight knob scales.

4.6.6.20 Resolution, on axis. - Position Slide #10 of Lamp "K" such that Filter "A" is in the optical line-of-sight. This will provide the projected resolution target of 22 L-pr/mn with a background luminance of 1 to 5 x $10^{-3}$ ft-1 and a contrast of 2.0 or greater. View through the elbow assembly eyepiece with the unaided eye and obtain best focus of the resolution target, by means of the eyepiece diopter adjustment ring. The resolution target shall have the detectable line structure and proper line count when observed through the eyepiece. Perform the above test with Filter "C" positioned in the optical line-of-sight. This will provide the projected resolution of 4 L-pr/mm with a background luminance of 6 x $10^{-5}$ ft-1 and a contrast of 2,0 or greater. Resolution, on axis, of the elbow assembly shall conform to the requirements specified in 3.6.10. If assembly fails this test, perform test specified 4.6.6.20.1.

CAUTION: The image tube must be erased (completely discharged), the observer dark adapted for not less than 15 minutes before performing the resolution tests.

4.6.6.20.1 Resolution, on axis - (failed assembly). -

a. Remove image intensifier tube from failed assembly. Assemble tube into a production Elbow Assembly, 11733401 that meets the requirements and calibration instructions set forth on Drawing D11738130.

b. Perform resolution test on calibrated-production elbow as set forth in 4.6.6.20. If the assembly passes the resolution test, the production elbow from which the tube was removed, shall be rejected. If the assembly fails resolution test, reject tube.

4.6.6.21 Luminance gain.- Position Slide #10 of Lamp "K" such that Filter "B" is in the otical line-of-sight. This will provide target liminance of 5 x $10^{-4}$ ft-1. Perform this test utilizing the photometer specified' in 4.6.2.3. The photometer shall be eyecorrected and positioned such that it intercepts the line-of-sight from the elbow assembly eyepiece. The eye-corrected photometer shall be used to measure luminance gain by comparing the luminance seen by the elbow assembly with the luminance at the elbow assembly eyepiece. During the test, the gain control knob shall be turned fully clockwise. The gain is the ratio of the luminance at the eyepiece divided by the target luminance. Luminance gain shall be not less than the value specified in 3.6.11.

A.6.6.22 Luminance gain control.- This test shall be performed during the luminance gain test in 4.6.6.21. Rotate the gain control knob clockwise from the full counterclockwise on position specified in 3.6.11.1. The gain shall increase in accordance with the requirement of 3.6.11.1.



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4.6.6.23 Field of view.- While observing through the elbow assembly the 110 milliradian marks on the target image shall be visable to meet the requirements of 3.6.12.

4.6.6.24 Projected reticle pattern.

4.6.6.24.1 Reticle lmage.- The elbow assembly reticle projector and image tube shall be energized by the application of .18 VDC (reference Drawing F11733401). The image of the projected reticle pattern image shall be well defined and clearly visible when viewed with the unaided eye to insure compliance with 3.6.13.1a and 3.6.13.1b.

4.6.6.24.2 Reticle accuracy.- The angular value between the bore-sight cross and maximum range graduation of the projected reticle pattern shall meet the requirement of 3.6.13.2 when measured on the target “image”.

4.6.6.25 Cleanliness, dirt.- The reticle projector lamp and the image tube shall be energized during this test. Prior to performing the test, the image tube shall be permitted a stabilization period of 15 seconds. Subsequent to the tube stabilization period, obtain best focus of the phosphor grain on the screen of the image tube by adjustment of the eyepiece diopter knob. The screen surface of the image tube shall then be visually inspected for cleanliness in accordance with 3.7b and shall conform to the requirements specified therein. “Cold ” and “Hot” spots on the screen surface of the image tube shall not be considered dirt.

4.6.7 Cleanliness, defects.- Remove the elbow assembly from the testing equipment. The reticle projector lamp and the image tube shall be deenergized for this test. With two 15 satt desk fluorescent lamps held close to the objective window and eyelens, view by reflected light the glass components of the body assembly utilizing the technique of shadowing to determine compliance with the cleanliness requirements of 3.7a.

CAUTION: Prolonged exposure of the image tube to direct rays of white light shall be avoided.

4.6.8 Sealing.- This test shall be performed utilizing a pressure gage capable of reading between 0.25 to 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check the internal pressure of the elbow assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the elbow assembly by means of the dew point” tester. Interior atmosphere of the elbow assembly shall conform to the requirements of 3.8. Upon satisfactory completion of this portion of the test, internal equipment conforming to 4.5.2.3.1. Pressurize the elbow assembly with dry nitrogen gas, havign a dew point at least as low as minus 25 degrees Fahrenheit, to 5 pound per square inch (psi), in accordance With the procedure for internal testing on Drawing F8565556.



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Retain the applied pressure in the elbow assembly for the the duration specified in 3.8. During this time period, the boresight knobs and the diopter adjustment shall be rotated to their clockwise and counterclockwise stops. At the end of this time period there shall be no evidence of leakage as indicated on the pressure gage of the-testing equipment, to determine compliance with 3.8. Upon satisfactory completion of the sealing test, the elbow assembly shall then be flushed with the dry nitrogen gas specified herein. The interior atmosphere of the elbow assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as minus 25 degrees Fahrenheit. When this atmosphere is attained within the elbow assembly, the elbow assembly shall conform to the requirement of 3.8.

NOTE: The Battery Ground Cap 11733479 and Cap 10515991 shall be re moved from the elbow assembly during the sealing test.

4.6.9 Reliability demonstration test.- A reliability demonstration test shall be conducted as part of first article test to assure compliance with the requirement of paragraph 3.9 at a 70% confidence level. A minimum of 25 hours on each sample elbow assembly (sample size 3) shall be accumulated after/during exposure to each nonoperating/environment of this specification. Additional hours are required at 75° ± 15°F in order to accumulate a minimum of 800 hours on each sample. The sample elbow assembly shall be tested until the total operating time (2408 hours minimum) satisfies the following relationship or until 6 failures have occurred or 14,000 hours (maximum) have been accomplished

$$2000 \leq \frac{2T}{\chi^2_{.30,\ 2(r+1)}}$$

T = Total test hours

$X^2$. 30, 2 (r+l) = Chi Square lower 70% confidence level with 2(r+l) degrees of freedom

r = Number of chargeable failures,

4.6.9.1 Reliability data.- Data collected will consist of all equipment malfunctions, number of hour-s completed when these malfunctions occurred, nature and cause of malfunction, time required to correct malfunction, and required corrective action.

4.6.9.2 Reliability failure definition.- Any malfunction which renders the elbow assembly inoperative or operative outside the limits of this specification. In determination of chargeable failure, the following clarification shall apply:

a. If, upon failure, analysis (to determine the cause of mal-function), an assignable cause (related to manufacturing process, procedure, basic design weakness, etc.) is determined as the basis of the malfunction, the malfunction will not be chargeable as a reliability failure; however,



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the correction to the manufacturing process or procedure (et al) shall be accomplished and. the testing, resumed at the environment at which the malfunction occured.

b. If upon failure analysis no assignable cause Is evident and the malfunction is considered random in nature, the malfunction shall be a "chargeable" reliability failure and as such, must be consider-in the above demonstration test equation (4.6.9).

4.6.10 Interchangeability. - This test shall be performed utilizing maximum and minimum interchangeability gages conforming to Drawings F11729724 and F11732100 respectively, to detetermie compliance With 3.2.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11733401. The level of protection shall-be as specified in the procurement document.

## 6. NOTES

6.1 Intended use.- The Elbow Assembly: 11733401, is a module unit of the Tank Commander's Periscope: M36E1 or the Tank Gunner's. Periscope: M32E1 and M35El primarily used in the fire control system of the 105 Iron, Gun Tank: M60 Series.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable packaging data sheet (see 5.1).
d. Applicable stock number.
e. Provisions for First Article Testing.

Custodian:

Army - MU

Preparing Activity:

Army-MU

Project No. 1240-A672



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-E-48441A(MU) Elbow Assembly, Passive: 11733401

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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